Group Project: Part 1

Group Members:
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Question 1: What is the myth or question you want to explore?
The question that we would like to explore is: How willing are students to switch from in-person to online classes during extreme weather conditions?

Question 2: Restate your myth or question as hypotheses
H0 (Null Hypothesis): There is no effect of weather conditions on students' willingness to switch from in-person to online classes.
HA (Alternative Hypothesis): Weather conditions have a significant effect on students' willingness to switch from in-person to online classes.

Question 3: What is your population, sample, parameter, and statistic?
Population: All university students who attend in-person classes.
Sample: Around 30–40 Stevens Institute of Technology students who respond to the online survey shared through social media platforms.
Parameter: The true average willingness of all university students to switch to online classes during extreme weather.
Statistic: The sample mean willingness score calculated from the collected survey responses.

Question 4: What kind of study is it?
This is an observational study because it involves collecting survey responses without manipulating any variables. We will not perform any random assignment or experimental control. Additionally, data will be collected based on students' self-reported willingness to switch to online classes during extreme weather.

Question 5: What are your dependent and independent variables?
Our independent variable is the weather condition, which can vary from mild to extreme. On the other hand, the dependent variable is students' willingness to switch from in-person to online classes during extreme weather.

Question 6: For each variable, describe the scale of measurement.
The independent variable, weather condition, is a categorical variable measured on a nominal scale with values such as mild, moderate, and severe. In contrast, the dependent variable, students' willingness to switch to online classes, is measured on an ordinal scale using a Likert scale from 1 (not at all willing) to 5 (very willing).

Question 7: How will you collect your data?

We will collect data through an online survey created using Google Forms. We will share the survey link on Facebook and WhatsApp groups for Stevens Institute of Technology students. Around 30–40 students are expected to respond voluntarily, providing anonymous answers about their willingness to switch to online classes during extreme weather conditions.

Question 8: What are potential sources of bias that could affect your results?

The potential sources of biases are as follows:

a. Voluntary response bias – only students who choose to participate may respond, which might not represent all students' opinions.
b. Sampling bias – only students active on Facebook or WhatsApp groups can access the survey link.
c. Nonresponse bias – students who are not comfortable with accessing the digital form may choose not to participate.
d. Interpretation bias – students may have different perceptions of what qualifies as "extreme" weather.

Question 9: What do you expect to find, and why will your results be valuable?

We expect to find that students are more willing to switch from in-person to online classes during extreme weather conditions due to safety concerns and convenience. Also, students with reliable internet access and a comfortable home environment are likely to show higher willingness to switch.

Finally, we think these findings will help universities to improve class flexibility, support student safety, and maintain learning continuity during extreme weather events.